

CHANTYRYA, Russian Federation

WMO: 239290

Lat: 60.3216N

Lon: 64.2165E

Elev: 53

StdP: 100.69

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -33.6	(d) -39.0	(e) 0.1	(f) -36.2	(g) -36.1	(h) 0.1	(i) -33.6	(j) 7.3	(k) -17.4	(l) 6.7	(m) -12.7	(n) 1.1	(o) 310	(p) 0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 29.4	(d) 20.0	(e) 27.6	(f) 18.9	(g) 25.5	(h) 17.9	(i) 21.1	(j) 27.4	(k) 20.0	(l) 25.7	(m) 18.9	(n) 24.0	(o) 2.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.8	23.5	17.8	12.8	23.0	16.7	11.9	22.0	61.3	27.3	57.4	25.9	53.8	24.2	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.9	(b) 6.7	(c) 6.0	(e) -40.2	(f) 32.4	(g) 3.9	(h) 1.2	(i) -43.0	(j) 33.2	(k) -45.3	(l) 34.0	(m) -47.5	(n) 34.6	(o) -50.3	(p) 35.5		
(a) 7.9	(b) 6.7	(c) 6.0	(e) -40.3	(f) 22.1	(g) 3.8	(h) 1.1	(i) -43.0	(j) 22.9	(k) -45.3	(l) 23.6	(m) -47.5	(n) 24.2	(o) -50.2	(p) 25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-18.2	-15.6	-6.1	2.0	9.2	15.8	18.1	14.8	9.0	1.5	-8.5	-14.3		
	DBStd	13.95	8.77	8.36	6.24	5.71	5.22	4.80	4.15	4.13	4.19	5.23	8.04	8.73		
	HDD10.0	4068	873	718	499	243	81	9	1	6	67	266	554	753		
	HDD18.3	6518	1132	951	757	491	286	103	57	126	280	521	804	1011		
	CDD10.0	690	0	0	0	3	57	183	252	154	38	3	0	0		
	CDD18.3	99	0	0	0	0	3	28	51	16	1	0	0	0		
	CDH23.3	942	0	0	0	0	53	252	489	141	7	0	0	0		
	CDH26.7	247	0	0	0	0	9	65	134	38	1	0	0	0		
(14) Wind	WSAvg	2.7	2.4	2.4	3.0	3.0	3.2	3.0	2.4	2.5	2.7	2.8	2.7	2.5		
(15) Precipitation	PrecAvg	406	22	18	19	24	33	53	54	56	44	35	28	23		
	PrecMax	521	47	55	38	44	93	133	98	87	100	74	57	41		
	PrecMin	231	6	4	2	0	8	6	14	19	14	6	5	9		
	PrecStd	70	9	10	9	12	18	28	25	19	24	18	13	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	2.5	8.5	17.8	28.1	31.0	31.6	30.5	24.3	16.6	6.5	2.3		
		MCWB	0.3	0.3	3.6	10.2	16.9	20.6	21.1	20.5	16.5	11.2	4.6	0.7		
	2%	DB	-1.8	0.4	5.6	14.4	24.8	28.6	29.7	27.2	20.8	13.6	4.4	0.7		
		MCWB	-2.7	-1.0	1.6	8.4	15.1	19.4	20.3	19.3	14.8	9.5	3.2	-0.4		
	5%	DB	-4.1	-1.8	3.6	12.3	21.7	26.4	28.1	24.3	17.9	11.1	2.5	-1.0		
		MCWB	-4.7	-3.0	0.6	7.1	13.6	18.1	19.4	17.7	13.2	8.2	1.3	-1.9		
	10%	DB	-6.3	-4.1	1.9	10.2	18.7	24.0	26.2	22.0	15.4	8.8	0.7	-3.2		
		MCWB	-6.9	-5.3	-0.4	5.6	12.2	17.0	18.7	16.6	12.0	6.8	-0.4	-4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.8	3.8	11.5	17.9	21.7	22.5	21.5	17.3	11.7	5.0	1.0		
		MCDB	0.8	2.0	8.0	16.0	26.6	28.7	28.9	28.6	22.7	15.1	6.3	1.9		
	2%	WB	-2.8	-0.9	2.1	9.2	15.8	20.1	21.5	19.8	15.2	10.1	3.2	-0.4		
		MCDB	-2.0	0.5	4.7	13.7	23.2	27.0	27.7	24.9	19.5	12.8	4.3	0.7		
	5%	WB	-4.7	-2.9	0.9	7.4	14.2	18.9	20.6	18.6	13.7	8.3	1.4	-1.9		
		MCDB	-4.2	-1.8	3.2	11.5	20.6	24.9	26.4	23.2	17.1	10.8	2.3	-1.0		
	10%	WB	-6.9	-5.3	-0.1	5.8	12.6	17.8	19.5	17.4	12.4	6.8	-0.1	-4.0		
		MCDB	-6.2	-4.2	1.7	9.5	18.2	22.9	24.7	21.2	15.1	8.7	0.7	-3.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	9.0	9.3	9.3	10.2	9.5	10.0	9.1	7.7	5.9	6.2	6.9		
		MCDBR	7.3	7.3	9.2	12.2	14.9	13.0	13.3	12.7	11.2	9.2	5.5	6.7		
		MCWBR	7.0	6.5	6.5	7.6	7.9	5.8	5.5	5.6	5.8	6.0	4.7	6.3		
		MCDBR	7.2	7.0	8.2	11.2	13.6	11.5	11.7	11.2	10.1	8.7	5.1	6.7		
	5% WB	MCWBR	7.0	6.3	5.9	7.5	7.6	5.5	5.4	5.5	5.7	5.9	4.7	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.301	0.353	0.388	0.391	0.398	0.414	0.415	0.362	0.328	0.279	0.249		
	taud		1.999	2.031	1.981	2.085	2.295	2.342	2.296	2.259	2.378	2.308	2.093	1.946		
	Ebn at Noon		461	660	740	796	833	832	805	768	754	646	441	310		
	Edh at Noon		47	83	121	132	116	113	116	111	83	64	43	32		
(44) All-Sky Solar Radiation	RadAvg		0.45	1.35	2.88	4.19	5.03	5.36	5.29	3.85	2.30	1.15	0.44	0.23		
	RadStd		0.03	0.09	0.17	0.45	0.41	0.44	0.47	0.36	0.24	0.14	0.08	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	-1.22	-0.97	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page